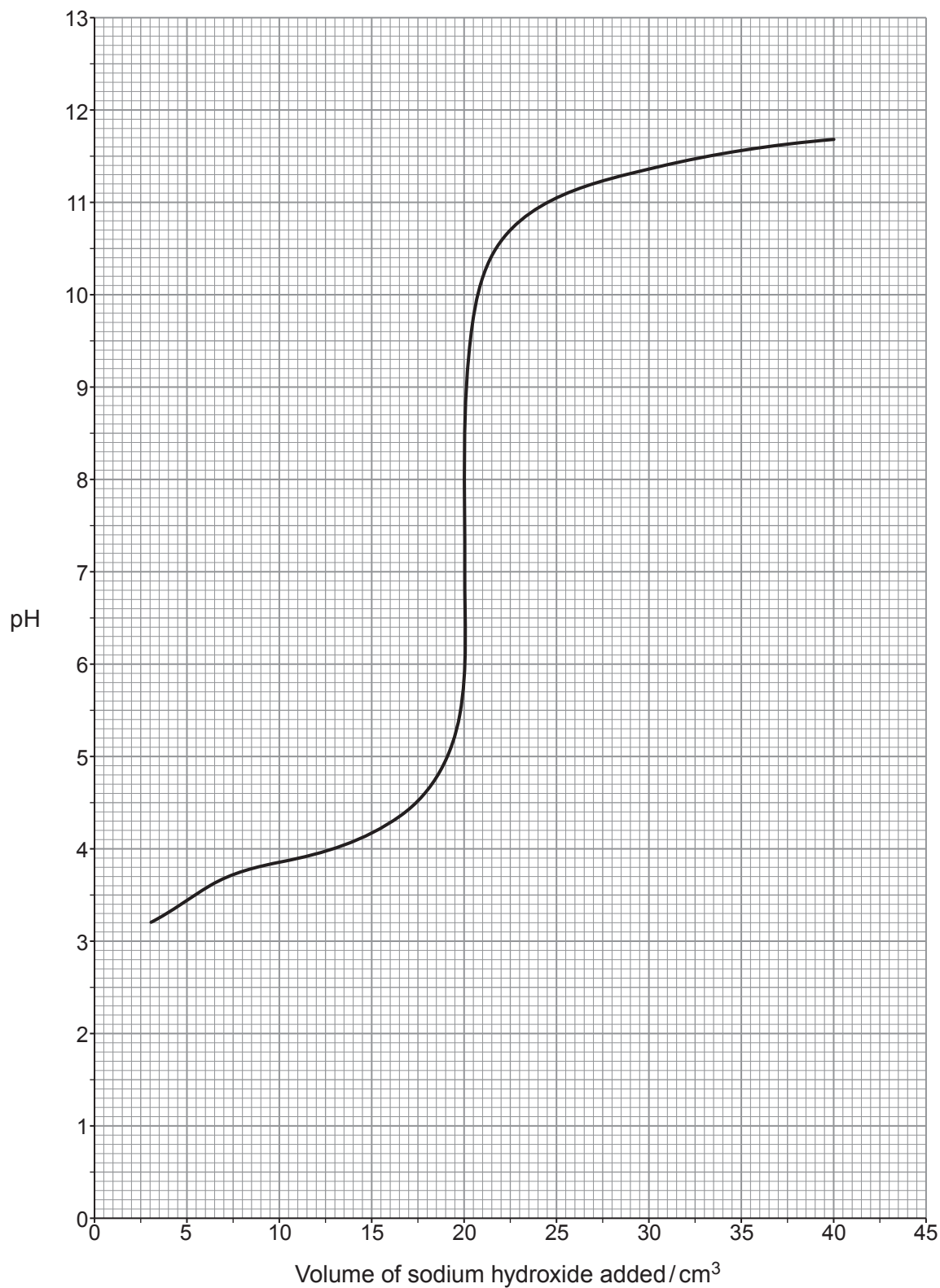


10. A student used a solution of weak acid HA in a titration. A 25.0 cm^3 sample of the acid was titrated against a sodium hydroxide solution of concentration 0.250 mol dm^{-3} giving the pH titration curve shown. The initial pH value is missing from the graph.



- (a) Several indicators are listed in the table.

Identify **all** the indicator(s) which are appropriate for this titration. Give a reason for your choice(s).

[2]

Indicator	pH range
methyl orange	3.1 - 4.4
bromocresol purple	5.2 - 6.8
bromothymol blue	6.0 - 7.6
naphtholphthalein	7.3 - 8.7
cresolphthalein	8.2 - 10.1

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- (b) Calculate the concentration of the HA solution.

[2]

Concentration = mol dm^{-3}



- (c) Find the pH of the mixture after addition of 10.0 cm^3 of sodium hydroxide solution and hence calculate the initial pH of the HA solution. [4]

You **must** show your working.

pH after addition of 10.0 cm^3 =

Initial pH =

- (d) HA is a weak acid and so it can be used to make a buffer solution.

Suggest **one** important use for buffer solutions. [1]

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- (e) Some data regarding the dissociation of another weak acid, HX, at two different temperatures are given in the table.



	HX at 298 K	HX at temperature T
Percentage dissociated	3.9	3.4
ΔG for acid dissociation / kJ mol^{-1}	15.9	16.7
ΔH for acid dissociation / kJ mol^{-1}	-10.0	
ΔS for acid dissociation / $\text{J K}^{-1} \text{mol}^{-1}$	-87.0	

Find temperature T and hence explain in terms of equilibrium why the percentage dissociation is different at this temperature.

[3]

$T = \dots\dots\dots \text{K}$

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